

The cleaning

1. Understand the dataset

- Load the Excel file.
- Check number of rows and columns.
- Identify data types.
- Understand what each column represents.

2. Check data quality

Look for:

- Missing values
- Duplicate rows
- Wrong data types
- Inconsistent text
- Invalid values
- Outliers
- Impossible values

3. Clean text columns

Examples:

- Remove leading/trailing spaces.
- Standardize blank values.
- Convert values such as "NA", "N/A", "null" into missing values.

4. Validate numerical columns

Check whether values make sense according to the variable.

For example:

Column	Validation
Age	Must be within a reasonable student-age range
Attendance Rate	0–100%
Exam Score	0–100
Stress Level	0–10
Sleep Hours	Reasonable range
GPA	Valid GPA range
Course Difficulty	1–5

5. Handle missing values

Choose the method based on the column:

- Numerical: median/appropriate statistical method
- Categorical: mode or another justified method
- Don't blindly fill everything with zero.

6. Handle duplicates

Identify exact duplicate records and remove them when they're genuinely duplicate observations.

7. Handle outliers

Use methods such as:

- IQR
- Z-score
- Domain/business rules

Important: An outlier isn't automatically an error. You should investigate it before deleting it.

8. Check consistency

Look for logical relationships between columns.

For example:

- Attendance shouldn't be >100%.
- Scores shouldn't be negative.
- Number of failed courses shouldn't be negative.
- GPA should be compatible with its defined scale.

9. Recheck the cleaned dataset

After cleaning:

- Missing values → check again
- Duplicates → check again
- Invalid values → check again
- Data types → check again
- Row/column counts → verify

10. Export the cleaned dataset

Finally:

Raw Data → Data Quality Check → Cleaning → Validation → Clean Excel File

For the nulls

The first col(age): For the Age column in your dataset, I handled the null values using median imputation.

Specifically:

1. I identified the missing values in age.
2. I checked the valid range for age.
3. For values outside the reasonable range, I treated them as missing.

4. I calculated the median age from the valid records.
5. I replaced the missing age values with that median.
6. After cleaning, there were 0 missing values in age.

Why median?

For an age variable, median is usually safer than mean because it is less affected by unusual values

So The null in table from col 1 to 41:

Three important decisions

For the last three columns, I would not fill the nulls with median/mode.

final_score, final_grade, and pass_fail are outcome/target variables. If we're going to analyze student performance, inventing their final result could seriously affect the analysis.

So I'd keep those missing values as Unknown/Missing or exclude those records only when performing analyses that require the outcome.

Our agreed cleaning plan so far

1. age → Median
2. gender → Unknown  *your first requested edit*
3. parent_income → Median
4. num_siblings → Median
5. family_support → Median
6. commute_time_min → Median

7. part_time_job → Unknown

8. previous_gpa → Median

9. num_failed_courses → Median

10. high_school_grade → Median

11–38. Numerical/ordinal columns → Median

11. course_type → Unknown

12. final_score → Don't impute

13. final_grade → Don't impute

14. pass_fail → Don't impute

I have not changed the Excel file. This is just the cleaning plan we're agreeing on first